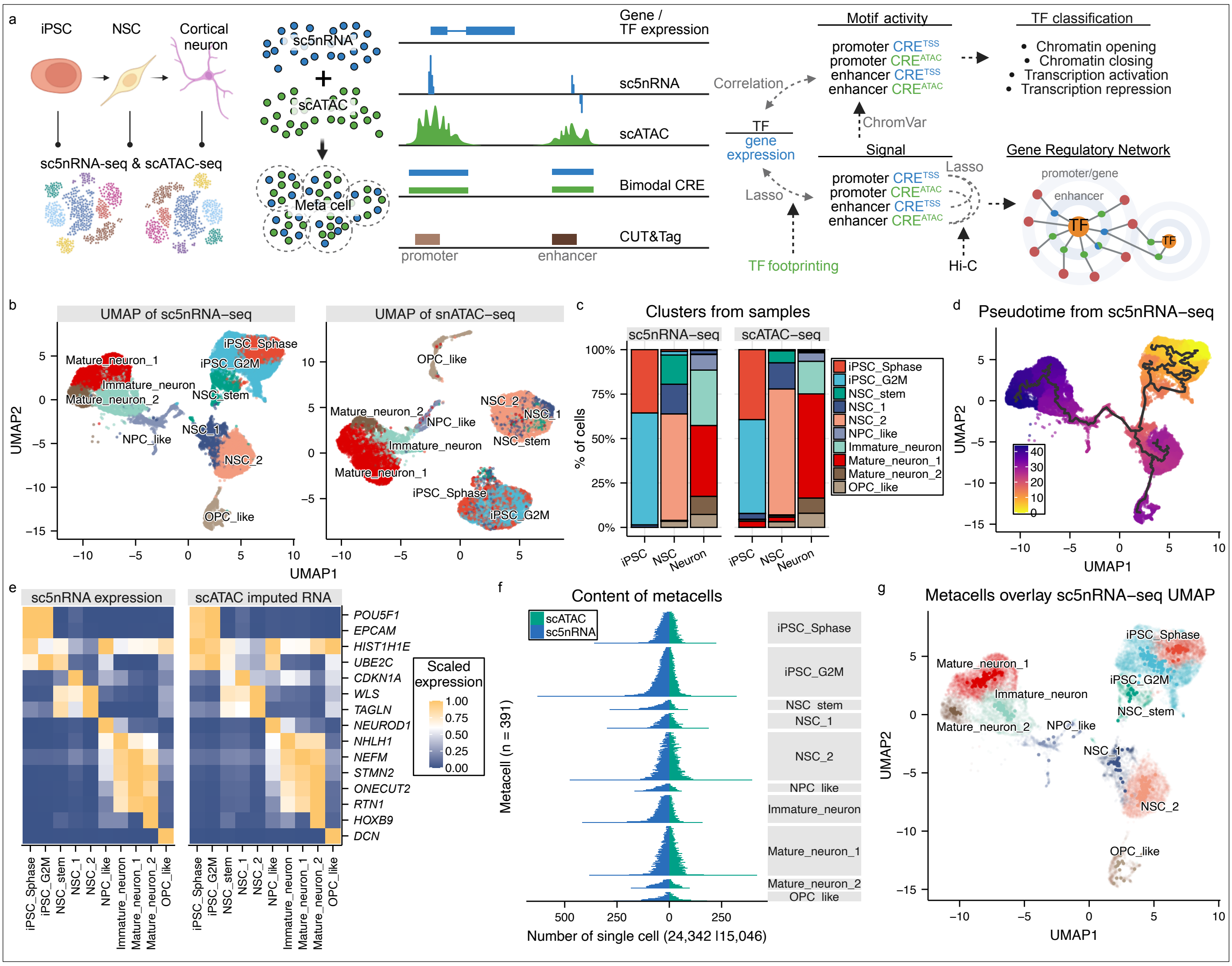

Figure 1 | Integration of sc5nRNA-seq and scATAC-seq.

a, Schematic of the study design illustrating the integration of sc5nRNA-seq and scATAC-seq across a differentiation time course from iPSC to cortical neuron. The workflow identifies *cis*-regulatory elements (CREs) which were quantified with bimodal signals (chromatin accessibility and transcription). These CREs were classified as enhancer-like or promoter-like based on CUT&Tag-derived chromatin states. Global transcription factor (TF) motif activity was quantified by ChromVAR while TF-enhancer-promoter gene regulatory network (GRN) was established via Lasso-based model. **b**, UMAP visualizations on single-cells according to the gene expression (sc5nRNA-seq) and chromatin accessibility (scATAC-seq). **c**, Cell clustering and relative proportions of the three samples across both modalities. **d**, Pseudotime trajectory defined by sc5nRNA-seq data. **e**, Scaled expression level of selected marker genes across 10 cell clusters, comparing gene expression from sc5nRNA-seq and the imputed gene activity scores from scATAC-seq. **f**, Number of single-cells from each modality contributing to individual metacells per cluster. **g**, UMAP of metacells based on sc5nRNA-seq signal, with the single-cell distribution shown underneath.